Recombinant Human Hyaluronidase PH20 Does Not Stimulate an Acute Inflammatory Response and Inhibits Lipopolysaccharide-Induced Neutrophil Recruitment in the Air Pouch Model of Inflammation

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Hyaluronidase (Hyal) and low m.w. hyaluronan (LMW HA) fragments have been widely reported to stimulate the innate immune response. However, most hyaluronidases used were purified from animal tissues (e.g., bovine testis Hyal [BTH]), and contain endotoxin and other unrelated proteins. We tested a highly purified recombinant human Hyal (rHuPH20) and endotoxin-free HA fragments from M_r 5,000 to 1,500,000 in the rodent air pouch model of inflammation to determine their potential for stimulation of the innate immune response. Exogenous LMW HA fragments (average M_r 200,000) failed to induce either cytokine/chemokine production or neutrophil infiltration into the air pouch. Challenging the air pouch with LPS or BTH stimulated production of cytokines and chemokines but rHuPH20 did not, suggesting that neither PH20 nor generation of LMW HA fragments in situ stimulates cytokine and chemokine production. LPS and BTH also induced neutrophil infiltration into the air pouch, which was not observed with rHuPH20 treatment. Endotoxin-depleted BTH had much reduced proinflammatory activity, suggesting that the difference in inflammatory responses between rHuPH20 and BTH is likely due to endotoxin contaminants in BTH. When rHuPH20 was dosed with LPS, the induction of cytokines and chemokines was the same as LPS alone, but neutrophil infiltration was inhibited, likely by interrupting HA–CD44 interaction. Our results indicate that neither rHuPH20 nor its directly generated HA catabolites have inflammatory properties in the air pouch model, and rHuPH20 can instead inhibit some aspects of inflammation, such as neutrophil infiltration into the air pouch. ***The Journal of Immunology*****, 2014, 192: 5285–5295.**

Hyaluronan or hyaluronic acid (HA) is a nonsulfated glycosaminoglycan composed of repeating polymeric disaccharides of D-glucuronic acid and *N*-acetyl-D-glucosamine, and is a component in extracellular matrix. Despite its simple structure, HA has been reported to have a great number of biological functions, such as stimulation of cell proliferation and migration, morphogenesis, wound healing, inflammation, angiogenesis, and tumor growth (1–8). In humans, HA is abundant in several tissues, including skin, umbilical cord, synovial fluid, cartilage, and skeletal tissues (9). The metabolism of HA is dynamic, with one-third of total HA turned over daily (9, 10). HA is synthesized by HA synthases (HAS1, 2, or 3) and degraded primarily by hyaluronidase (Hyal) family members in animals (11). Hyals hydrolyze HA at the hexosaminidic (1→4) linkages between *N*-acetyl-D-glucosamine and D-glucuronic acid residues and release HA fragments as either oligosaccharides or larger polymers, depending on the extent of cleavage. There are six human Hyal genes identified, including HYAL1, 2, 3, 4, PH20/SPAM1, and pseudogene HYALP1 (12). Testicle-derived Hyal, containing PH20, has been used widely in research and a variety of clinical applications (13).

A number of reports suggest a role for HA in the regulation of the immune response. Under normal noninflammatory conditions, HA exists primarily as high m.w. (HMW) HA (M_r >1,000,000). Under stress or inflammatory conditions, low m.w. (LMW) HA (M_r <1,000,000) accumulates and has been reported to induce inflammatory responses in immune cells (14–21). Additional reports suggest that LMW HA has proinflammatory activity, whereas HMW HA has anti-inflammatory properties. Treatment with LMW HA leads to increased LPS-induced lung inflammation either directly or following priming with ozone, whereas treatment with HMW HA protects mice from liver injury by reducing proinflammatory cytokine production (22, 23). LMW HA fragments, generated by Hyals and reactive oxygen species under physiologic stress (such as tissue injury and inflammation), are thus hypothesized to function as immune regulators (2) and induce proinflammatory cytokines and chemokines in vitro via HA fragments interacting with TLR2 and TLR4 (24, 25). However, no direct interaction between LMW HA and TLR2 or TLR4 has been demonstrated, despite references in the literature (2, 26). In addition, the reagents (including Hyals and their LMW HA products) used from some sources may contain other inflammation-stimulating components, such as endotoxin and growth factors (14), which makes it less clear whether it is LMW HA or impurities in the tested reagents that contribute to the proinflammatory action. Recent evidence that supports skepticism in this area is

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Abbreviations used in this article: BTH, bovine testis hyaluronidase; HA, hyaluronan or hyaluronic acid; HMW, high m.w.; Hyal, hyaluronidase; KC, keratinocyte chemoattractant; LMW, low m.w.; LTA, lipoteichoic acid; rHuPH20, recombinant human hyaluronidase PH20; SEAP, secreted embryonic alkaline phosphatase.

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the observation that recombinant human hyaluronidase PH20 (rHuPH20), which degrades HMW to LMW HA fragments, did not stimulate IL-8 production in HUVECs (27). Other publications also showed that low endotoxin–containing HA fragments ranging from *M*r 1,000 to 1,500,000 did not stimulate IL-1α production in primary human macrophages (28), or NO production in the murine cell line RAW 264.7, rat splenocytes, or rat adherent differentiated primary macrophages (29). HA at both high and low m.w. reduced LPS-induced inflammation in fibroblast-like synovial cells (30). Ebid et al. (31) also showed that endotoxin contamination of bovine testis Hyal (BTH), rather than HA fragments generated by Hyal degradation, stimulated TLRs in mesangial cells and induced an inflammatory response, suggesting that HA fragments are not TLR ligands. Therefore, the current literature does not clearly demonstrate the role of HA fragments in modulating an inflammatory response.

A key feature of acute inflammation is neutrophil recruitment into the inflammatory site by cytokine- and chemokine-mediated chemotaxis, during which HA and CD44 play a role. The interaction between HA on the luminal endothelium and the HA receptor CD44 (present on neutrophils) mediates neutrophil-endothelial cell interactions, causes neutrophil adhesion and migration, and contributes to optimal leukocyte recruitment into tissues in response to proinflammatory stimulation (32, 33). CD44 on lymphocytes also binds HA on endothelium to mediate the primary adhesion/rolling of lymphocytes on vascular endothelial cells under conditions of physiologic shear stress, and this interaction is important for activated T cell extravasation into an inflamed site in vivo (34–36). It is therefore hypothesized that Hyal, by degrading and removing HA from tissue, can interfere with the HA–CD44 interaction and consequently reduce neutrophil recruitment into tissues during inflammation, provided that the Hyal used is free of proinflammatory contaminants.

The air pouch model of inflammation displays several features of the innate immune system since the immune cells that line the pouch exhibit similarities to the innate immune cell components of tissues (37). Following injection of inflammatory stimuli such as LPS, immune cells in the pouch are activated and secrete a variety of cytokines and chemokines, resulting in chemotactic migration of leukocytes into the pouch (38, 39). Because the air pouch is formed s.c., it is in close proximity to the skin, a major source of HMW HA (40). In the air pouch model, when PH20 is injected into the air pouch it can enzymatically cleave HMW HA into LMW HA, which may then exert its activity on the innate immune system in situ. In this study, we used the air pouch model to evaluate the impact of rHuPH20 and BTH (Hyal type I-S and IV-S) on acute inflammation by determining whether enzyme administration (with associated degradation of HMW HA) would stimulate production of cytokines and chemokines, as well as neutrophil infiltration into the air pouch cavity. Purified exogenous LMW HA fragments (M_r ~200,000) were also tested in the model.

Materials and Methods

Reagents

rHuPH20 was from Halozyme Therapeutics (San Diego, CA). LPS from *Escherichia coli* serotype O111:B4 and BTH Hyal type I-S and IV-S were from Sigma-Aldrich (St. Louis, MO). Endotoxin-free HA fragments at various m.w. were from Lifecore Biomedical (Chaska, MN). HEK-Blue hTLR2 and HEK-Blue hTLR4 cells were from Invivogen (San Diego, CA). Murine macrophage RAW 264.7 cells were from American Type Tissue Collection (Manassas, VA). Rat anti-mouse CD44 Ab (clone IM7) was from eBioscience (San Diego, CA). Six- to 7-wk-old male C57BL/6 mice were obtained from Taconic (Oxnard, CA), and were maintained in the animal facility of Halozyme Therapeutics. The animal use protocol was reviewed and approved by the Institutional Animal Care and Use Committee.

HEK-Blue reporter assay

HEK-Blue hTLR2 and hTLR4 cells were treated with HA with an average size of *M*r 5,000, 10,000, 20,000, 40,000, 60, 000, 100,000, 200,000, 500,000, or 1,500,000 at 100 μg/ml. PBS was used as negative control. Lipoteichoic acid (LTA) (50 μg/ml) was used as a positive control for HEK-Blue hTLR2, and LPS (0.1 μg/ml) was used as positive control for HEK-Blue hTLR4. Secreted embryonic alkaline phosphatase (SEAP) activity was detected by QUANTI-Blue reporter assay, and the final quantification was based on the measurement of OD650 in each well. Data are presented as fold induction of reporter activity using PBS-treated samples as reference.

Cell-based TLR2/TLR4 binding assay

HEK-Blue hTLR2 and hTLR4 cells were cultured on poly-L-lysine–coated Nunc LabTek II coverglass (Sigma-Aldrich) at 20,000 cells/well density for 24 h, followed by incubation for an additional 2 h with biotinylated Select-HA fragments (100 μg/ml) of *M*r 50,000, 250,000, or 1,000,000 (Lifecore Biomedical) in their normal culture media. Cells were then washed with 1× PBS three times and incubated with LPS Alexa Fluor 488 (10 μg/ml), streptavidin Alexa Fluor 594 (4 μg/ml), and Hoechst 33342 (0.5 μg/ml; Life Technologies) for 1 h in normal culture media. Subsequently, cells were washed again with 1× PBS and imaged on a fluorescent microscope with Metamorph image acquisition program. MDA-MB-468 cells characterized as expressing a high level of CD44 were processed similarly to confirm the binding capacity of biotinylated HA of *M*r 250,000 and 1,000,000.

ELISA for murine MIP-2

Mouse macrophage cell line RAW 264.7 was maintained in DMEM supplemented with 10% FBS and 4 mM GlutaGro (Corning Cellgro, Manassas, VA) in 5% CO_2 at 37°C. RAW 264.7 cells were seeded at 10,000 cells/well in 96-well plates and challenged with different reagents the next day. Culture supernatant was collected 4 h after the application of test articles and stored at 4°C. MIP-2 level in culture supernatants was determined with Mouse CXCL2/MIP-2 Quantikine ELISA Kit (R&D Systems, Minneapolis, MN). Data were presented as fold change with PBS-treated samples as reference.

Murine air pouch model

On days 0, 4, and 7, the animals were anesthetized with isoflurane and injected with 3 ml sterile air under the skin on the back. On day 10, 1 ml PBS was injected into the air pouch, alone or with indicated challenging reagents (including rHuPH20, HA, Hyal type I-S and IV-S, and LPS). At different time points after injection, the animals were anesthetized, and the pouch was washed with 1 ml PBS. The lavage fluid was cooled on ice, and 50 μl was used for cell counting by cellometer. The remaining fluids were centrifuged at 2,000 rpm for 5 min, and supernatants were stored in −80°C for cytokine and chemokine analysis. The cell pellets were resuspended in 500 μl PBS with 0.5% FBS at 4°C for FACS analysis.

ELISA-like HA assay and size determination

HA concentration in air pouch exudates and plasma was quantified using an enzyme-linked HA-binding protein sandwich assay (catalog number DY3614; R&D Systems), according to manufacturer's instructions. For HA size determination, air pouch exudates and plasma samples were treated with proteinase K, followed by HA extraction and resolved in 0.7% agarose gel (Tris-acetate-EDTA buffer). HA molecules were transferred to Hybond-N^+ membrane (Amersham Biosciences, Chandler, AZ) through a Southern blot transfer process in 100 mM Tris-acetate (pH 7.3). The Hybond-N^+ membrane was then processed following standard Western blot procedures with PBS + 0.1% Tween 20 as wash buffer, 2% dry milk in wash buffer as blocking reagent, 0.5 μg/ml biotinylated TSG6-Fc/dHep as primary Ab, and 0.2 μg/ml HRP-conjugated streptavidin (Jackson ImmunoResearch Laboratories, West Grove, PA) as secondary Ab.

Cytokine and chemokine analysis

The multiplex immunoassay to test cytokine and chemokine levels was conducted using EMD Millipore Mouse Cytokine/Chemokine Panel Kit (catalog number MCYTOMAG-70K; EMD Millipore, Temecula, CA) as described in the manufacturer's protocol.

Neutrophil analysis by FACScan

Immunofluorescence staining for murine neutrophils was conducted using anti-CD45 (clone 30-F11) and anti–Ly-6G (clone 1A8) Abs from BioLegend (San Diego, CA). Flow cytometric scanning of stained neutrophils

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was conducted using DIVA platform cytometer (BD Biosciences, San Jose, CA), and the data were analyzed by FlowJo Analysis Software (FlowJo, Ashland, OR).

Endotoxin assay and removal

Endotoxin level was determined with Endosafe PTS20 (Charles River Laboratories, San Diego, CA) following the manufacturer's instructions. The HA used in our study contained very low levels of endotoxin (<1 U/mg). Endotoxin in Hyal type I-S and IV-S was removed using the Triton X-114 extraction method (Thermo Fisher Scientific, Waltham, MA). Briefly, test reagent was mixed with Triton X-114 (1%) with gentle rocking at 4°C for 30 min, followed by 10 min incubation at 37°C. The mixture was then centrifuged at room temperature for 10 min at 10,000 × *g*, and the supernatant was transferred to a new tube for analysis.

Histochemical staining of HA, CD44, and neutrophils in skin patch of air pouch

The skin patches of air pouch were excised and fixed in 10% neutralized formalin for 48 h then embedded in paraffin. The paraffin-embedded skin patches were cut, dewaxed, and rehydrated in deionized water. Histochemical methods were used to stain HA in skin patches as described previously (41). For HA and CD44 coimmunofluorescence labeling, Ag retrieval was processed by heating slides in EDTA buffer (pH 8) at 100°C for 25 min. Slides were rinsed in PBS with Tween-20 (PBS-T), blocked with 2% normal goat serum in 2% PBS/BSA for 30 min, followed by incubation with HA-binding protein and anti-CD44 Ab (1:200 dilution; R&D Systems) for 2 h at room temperature. The sections were then incubated in Texas Red–tagged goat anti-rabbit IgG (1:200 dilution; Vector Laboratories, Burlingame, CA) for 1 h and mounted with ProLong Gold antifade reagent with DAPI (Invitrogen, San Diego, CA). Micrograph images were captured using a Zeiss Axioskop microscope coupled with an RT3 camera under a ×100 objective lens (Diagnostic Instruments, Sterling Heights, MI).

Protein purity determination

Protein samples (10 μg rHuPH20, 10 μg Hyal type I-S, and 10 μg Hyal type IV-S) were diluted in 15 μl 1× SDS sample loading buffer containing β-mercaptoethanol, resolved on 4–20% Mini-Protean TGX gel (Bio-Rad), stained with Instant Blue (Expedeon) for 2 h, and destained with deionized water for 1 h. Gel images were acquired using a Bio-Rad Chemi-Doc MP imaging system.

Statistical analysis

The results are presented as mean ± SD of the number of independent samples analyzed. Statistical significance was calculated using the Student two-tailed unpaired *t* test or ANOVA. $p < 0.05$ was considered significant.

Results

LMW HA does not bind or stimulate TLR2 or TLR4

To explore whether HA fragments can bind to TLR2 and TLR4 directly, biotinylated HA fragments of M_r 50,000, 250,000, or 1,000,000 were applied to cultures of HEK-Blue hTLR2 and HEK-Blue hTLR4 cells, and the putative interaction with cell surface was visualized by streptavidin conjugated with Alexa Fluor 594. Breast cancer cell line MDA-MB-468 expressing abundant CD44 and with demonstrated HA binding (42) was used as positive control for HA cell surface binding. Whereas both HA M_r 250,000 and 1,000,000 bound MDA-MB-468 cells, none of the HA species tested interacted with HEK-Blue hTLR2 or HEK-Blue hTLR4 cells. In addition, Alexa Fluor 488–conjugated LPS was able to bind both HEK-Blue hTLR2 and HEK-Blue hTLR4 cells (Fig. 1A) but not HEK-Blue null cells that do not express exogenous TLRs (data not shown). These data suggest that HA fragments do not interact with TLRs directly.

To determine whether HA fragments can induce hTLR2- or hTLR4-dependent signaling, a full range of HA (M_r 5,000–1,500,000) was tested in a TLR2- and TLR4-mediated reporter gene assay. LPS and LTA (as positive controls) dramatically stimulated TLR2- and TLR4-mediated reporter gene activity (Fig. 1B, 1C). None of the tested HA fragments showed stimulation of TLR2- or TLR4-mediated reporter gene activity, suggesting that HA fragments (including LMW HA) do not signal through TLR2 or TLR4 in this model system, although a rigorous response was observed with both LPS (TLR4 mediated) and LTA (TLR2 mediated). We further tested the full m.w. range of HA for

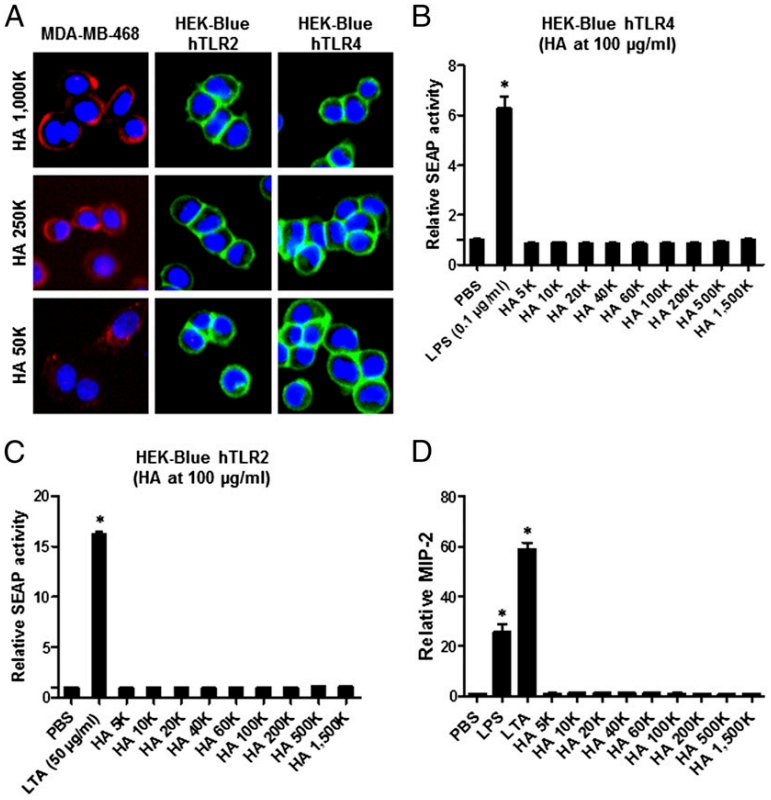

FIGURE 1. HA did not bind to TLR2 or TLR4, or stimulate TLR2- or TLR4-mediated NF-κB signal pathway or consequent MIP-2 production. (**A**) Binding of HA fragments (labeled with Alexa Fluor 594, red) and LPS (labeled with Alexa Fluor 488, green) to MDA-MB-468, HEK-blue TLR-2, and HEK-blue TLR-4 cells was presented. The labeled HA fragments and LPS were applied to the cells and the cells were imaged at original magnification ×100 by fluorescent microscopy. (**B** and **C**) Human TLR2- or TLR4-overexpressing HEK293 cells with NF-κB promoter–driven SEAP reporter gene coexpressed (Invitrogen) were used to test TLR-mediated NF-κB pathway stimulation. The SEAP reporter gene activity is shown, with LPS as the positive control for TLR4 (B) and LTA as the positive control for TLR2 (C). HA fragments at various m.w. (M_r 5,000–1,500,000; Lifecore Biomedical) were tested. (**D**) Murine macrophage cell Raw 264.7 were treated for 6 h with LPS (1 μg/ml), LTA (50 μg/ml), or HA fragments at various m.w. (100 μg/ml). Production of MIP-2 protein was shown. Results are mean readout or relative concentrations ± SD of triplicate cultures from one of three similar experiments. $^*p < 0.05$.

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stimulation of MIP-2 production in the murine Raw 264.7 macrophage cell line. We did not find stimulation of MIP-2 production by HA fragments from low (M_r 5,000) to high (M_r 1,500,000) m.w., although both LPS and LTA stimulated murine macrophages for MIP-2 production (Fig. 1D). Our data are consistent with recent publications reporting that LMW HA does not stimulate chemokine production in the RAW 264.7 cell line, rat splenocytes, rat adherent differentiated primary macrophages, or primary human macrophages (28, 29).

rHuPH20 or LMW HA do not stimulate production of cytokines/chemokines in the air pouch model

We investigated whether LMW HA or rHuPH20, which degrade HMW HA to LMW HA, can stimulate innate immune responses,

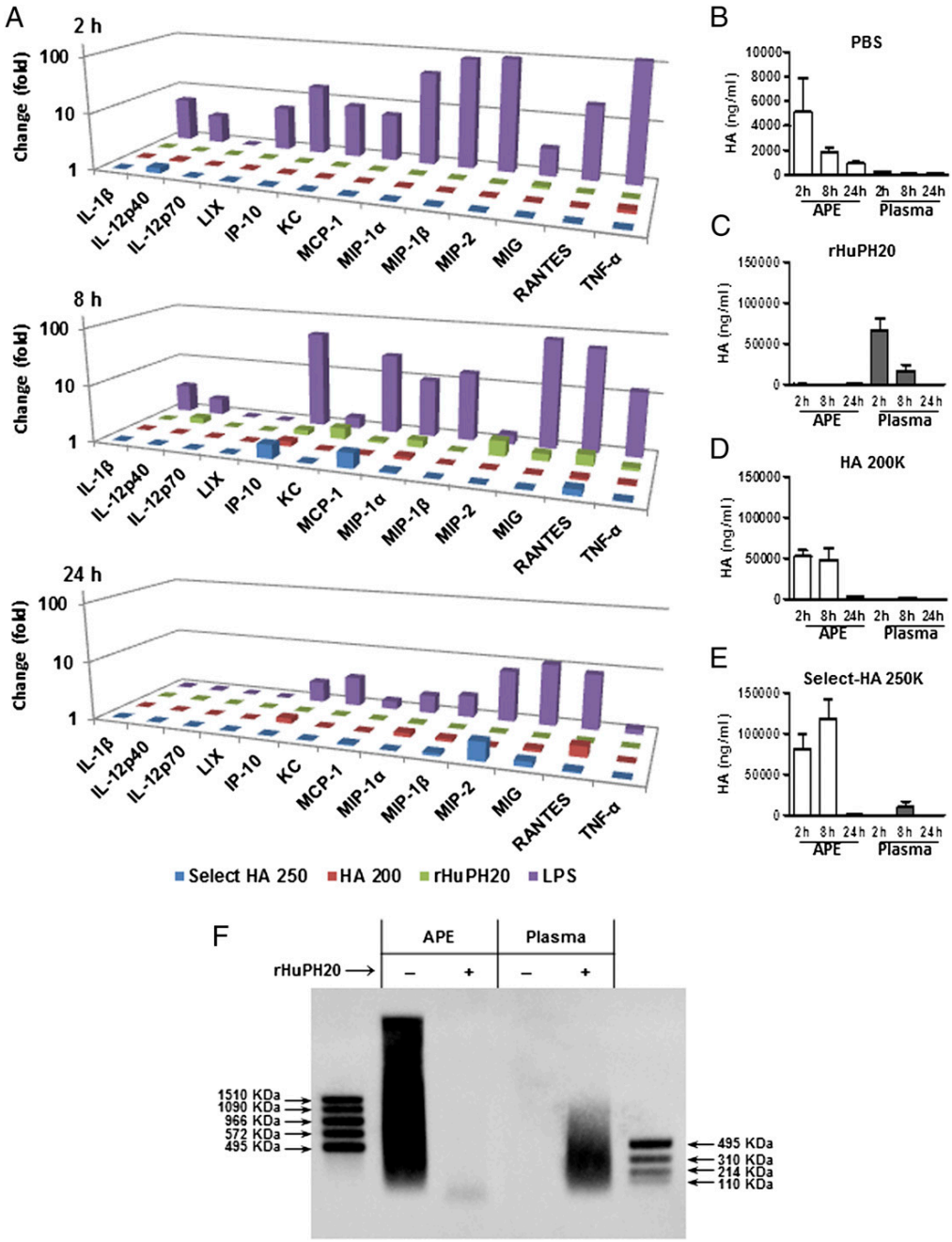

FIGURE 2. Neither LMW HA fragments nor rHuPH20 stimulated cytokines or chemokines in the air pouch. (**A**) The mouse air pouch was challenged with LPS (1 μg/ml), LMW HA fragments (Select-HA 250 and HA 200, 500 μg/ml), or rHuPH20 (20 μg/ml) for 2, 8, and 24 h. The air pouch exudates were collected, and the protein levels of cytokines and chemokines were determined by Luminex Multiplex assay. The protein level changes of cytokines or chemokines from baseline levels (PBS treated) are shown (n = 8). (**B**–**E**) The HA levels in the challenged air pouch, and plasma were determined by HA ELISA and shown for the groups injected with PBS [(B), the negative control], rHuPH20 (C), HA-200k (D), or Select-HA 250K (E). (**F**) rHuPH20 (20 μg) was injected into the air pouch. One hour postinjection, HA fragments in air pouch exudates and plasma were extracted, followed by agarose gel electrophoresis and blotted by TSG6-dHep-Fc. The HA level in plasma without rHuPH20 treatment is below detectable level of the TSG6-dHep-Fc blotting. Results are mean concentrations ± SD of eight replicates from one of three similar experiments.

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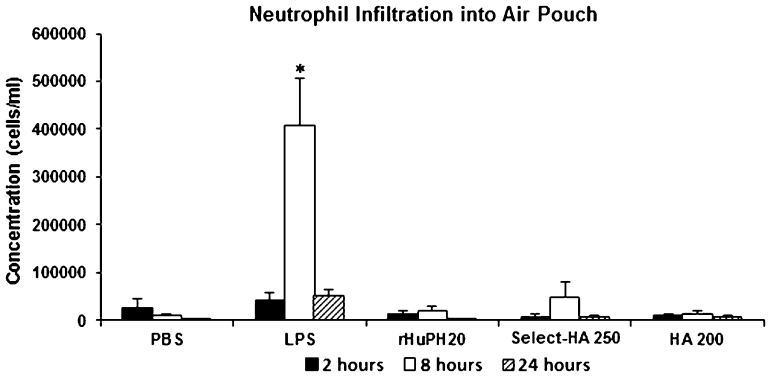

FIGURE 3. Neither LMW HA fragments nor rHuPH20 induced neutrophil influx into the air pouch. The mouse air pouch was challenged with LPS (1 μg/ml), LMW HA fragments (Select-HA 250 and HA 200, 500 μg/ml), or rHuPH20 (20 μg/ml) for 2, 8, and 24 h. Neutrophil density in the air pouch exudates was detected by FACS analysis using anti-CD45/anti–Ly-6G ($n = 8$). Results are mean concentrations ± SD of eight replicates from one of three similar experiments. $*p < 0.05$.

using the rodent air pouch model of inflammation (37). Protein levels of 13 inflammatory murine cytokines and chemokines in air pouch exudates and plasma were tested upon challenge with LMW HA, rHuPH20, or BTH in the air pouch.

Challenging the air pouch with LPS strongly stimulated production of IP-10, MCP-1, MIP-1α, MIP-1β, MIP-2, MIG, RANTES, and TNF-α in the air pouch exudates (Fig. 2A). The stimulation profiles were time dependent, with the peak stimulation occurring 2 h post dosing for MIP-1α, MIP-1β, MIP-2, and TNF-α, and at 8 h postdosing for IP-10, MCP-1, MIG, and RANTES. At 24 h postdosing, the levels of stimulated cytokines and chemokines returned to baseline. These data indicate that the rodent air pouch model has a sensitive and intact immune response to proinflammatory stimulation. In contrast to LPS treatment, LMW HA fragments (Select-HA 250 and HA 200) did not stimulate cytokine or chemokine production (Fig. 2A). Similar to LMW HA treatment, dosing of rHuPH20 did not stimulate production of any cytokine or chemokine in air pouch exudates (Fig. 2A).

LPS injected into the air pouch did not stimulate production of most cytokines and chemokines in plasma, except for keratinocyte chemoattractant (KC) with a mild induction at the 2-h time point. LMW HA and rHuPH20 also did not stimulate production of any of the tested cytokines or chemokines in plasma (data not shown). On the basis of these observations, neither LMW HA nor rHuPH20 stimulated inflammation locally in the air pouch or systemically in plasma.

To determine whether there was a significant level of HA exposure in the study animals, the samples from air pouch exudates and plasma were analyzed for HA concentration by an ELISA-like HA assay, and HA size was determined using agarose gel separation and probed with TSG6-dHep-Fc, a fusion protein with high-affinity HA binding (L. Huang, unpublished results). In air pouch exudates, the baseline HA concentration (PBS control animals) ranged between 1000 and 5000 ng/ml (Fig. 2B). Injection of 500 μg LMW HA fragments into the air pouch resulted in elevated HA levels from 50,000 to 120,000 ng/ml in air pouch exudates for up to 8 h (Fig. 2D, 2E). rHuPH20 treatment resulted in a strong reduction in HA (100–200 ng/ml; Fig. 2C). The reduction in HA concentration was observed at all time points within 24 h postdosing, indicating that rHuPH20 actively degraded HA in the air pouch as expected.

In plasma, baseline HA levels were low (100–200 ng/ml) in the control animals. rHuPH20 administration into the air pouch resulted in an increase of HA level in plasma, which was detected at 20,000–60,000 ng/ml up to 8 h postdosing, coincident with the reduction of HA level in the air pouch (Fig. 2E). Because rHuPH20 dosed into the air pouch did not lead to a detectable protein level of rHuPH20 in plasma (data not shown), the high level of HA that appeared in plasma likely came from HA in the air pouch mobilized by rHuPH20 degradation.

Degradation of HMW HA by rHuPH20 was confirmed by HA size analysis in air pouch exudates, with the appearance of LMW HA fragments in plasma (Fig. 2F). The control animals had HMW HA localized in the air pouch, but not in plasma, consistent with HA levels determined by ELISA shown in Fig. 2C. Following rHuPH20 treatment, the HMW HA band was reduced in the air

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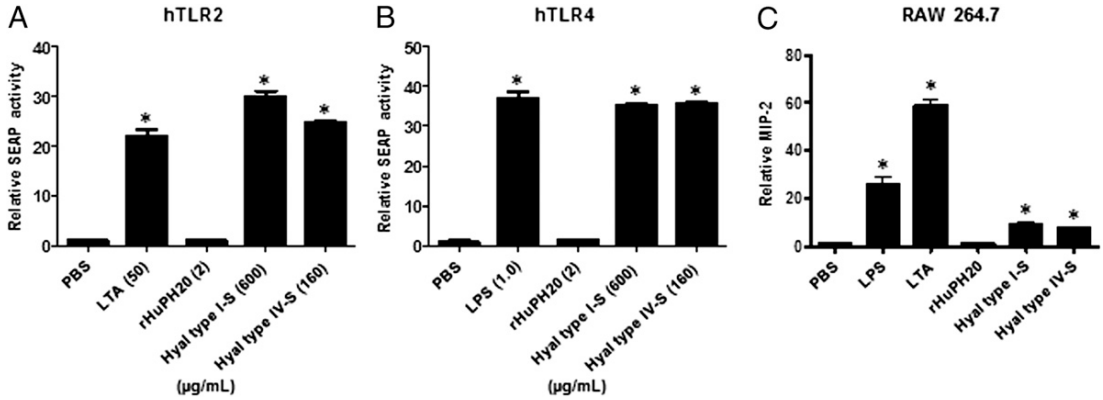

FIGURE 4. Hyal type I-S and IV-S are proinflammatory in vitro. (**A** and **B**) Human TLR2- or TLR4-overexpressing HEK293 cells with NF-κB promoter–driven SEAP reporter gene were used to test TLR-mediated NF-κB pathway stimulation. The SEAP reporter gene activity is shown for TLR2 (A) and TLR4 (*B*). rHuPH20 and Hyal type I-S and IV-S (Sigma-Aldrich) were tested. (**C**) Murine macrophage cells (Raw 264.7) were treated for 4.5 h with LPS (0.1 μg/ml), LTA (500 μg/ml), rHuPH20 (2 μg/ml), Hyal type I-S (600 μg/ml), and IV-S (160 μg/ml). The conditioned media were collected, and the production of MIP-2 protein was measured by ELISA. Results are mean readout or relative concentrations ± SD of triplicate cultures from one of three similar experiments. $*p < 0.05$.

pouch, suggesting an enzymatic effect of rHuPH20 on local HA metabolism in the air pouch. In plasma, rHuPH20 treatment resulted in an increase of LMW HA fragments (M_r 100,000–500,000), corresponding to HMW HA degradation in the air pouch.

Thus far, we have shown that dosing of LMW HA and rHuPH20 resulted in substantial levels of LMW HA exposure locally in the air pouch and systemically in plasma respectively for up to 8 h, but coincident stimulation of cytokine or chemokine production was not observed.

LMW HA or rHuPH20 does not induce neutrophil infiltration into the air pouch

We used Ab-based FACS analysis to detect neutrophil infiltration into the air pouch as a measure of the stimulation of the innate

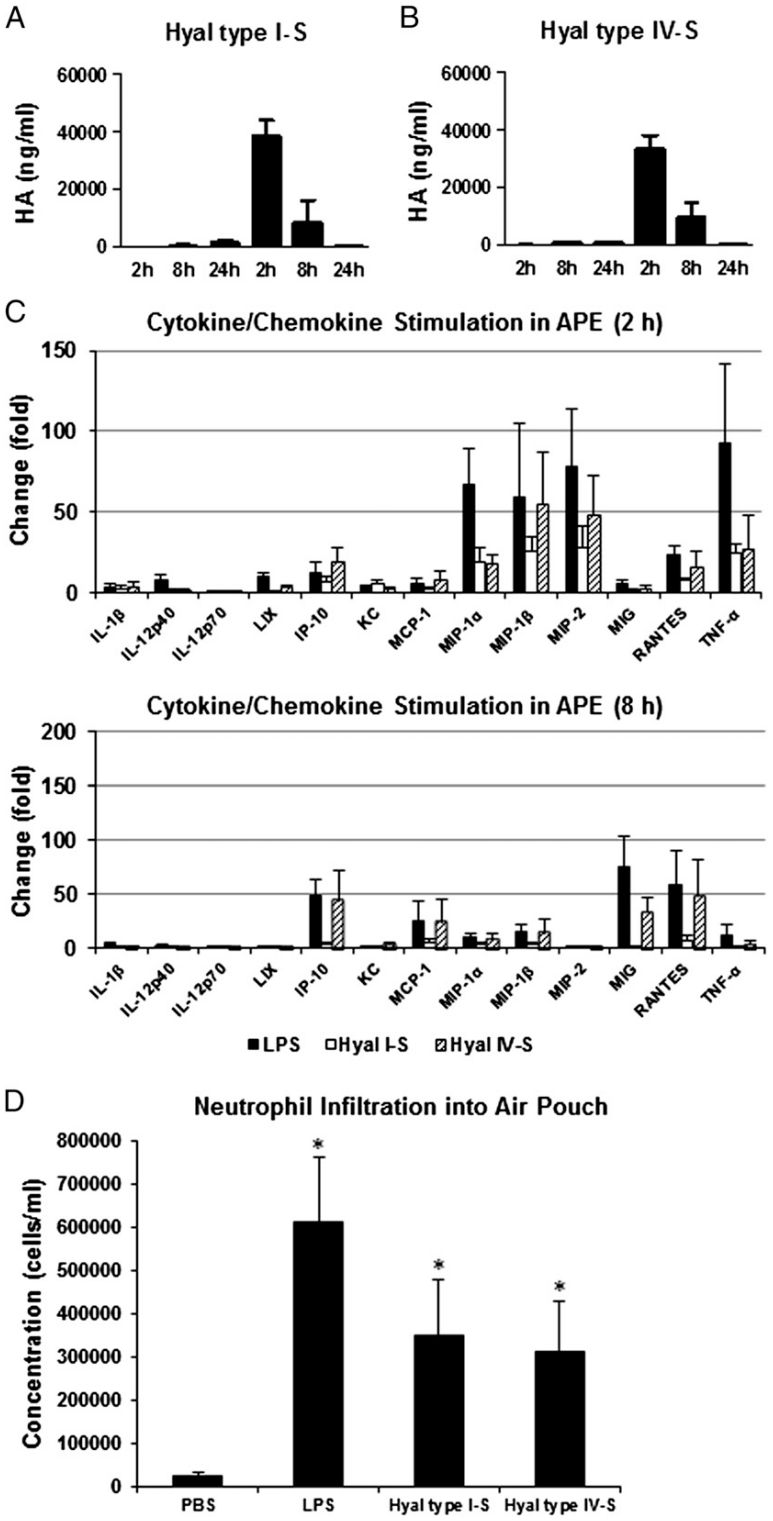

FIGURE 5. Hyal type I-S and IV-S are proinflammatory in vivo. The mouse air pouch was challenged with Hyal type I-S (6 mg/ml or 2400 U/ml) or Hyal type IV-S (1.6 mg/ml or 2400 U/ml) for 2 and 8 h. The air pouch exudates and plasma were collected. The HA levels in the challenged air pouch were determined by HA ELISA, and data are shown for the groups injected with Hyal type I-S (**A**) or IV-S (**B**). (**C**) The protein levels of cytokines and chemokines were determined by Luminex multiplex assay. Stimulation of the production of cytokines and chemokines in the air pouch are shown ($n = 8$). (**D**) Neutrophil density in the air pouch exudates was detected by FACS analysis using CD45/Ly-6G ($n = 8$). Results are mean readout or relative concentrations ± SD of eight replicates from one of three similar experiments. $^*p < 0.05$.

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immune response. At 8 h posttreatment, LPS induced a dramatic increase in the number of neutrophils in the air pouch, which decreased at 24 h posttreatment following the recession of cytokines and chemokines (Fig. 3). In contrast, none of the LMW HA fragments or rHuPH20 induced neutrophil infiltration, further confirming that LMW HA and rHuPH20 are not proinflammatory.

BTH preparations Hyal type I-S and IV-S are proinflammatory

Hyals, including Hyal type I-S and IV-S (BTH from Sigma-Aldrich), have been reported to stimulate the immune response, which has been attributed to the Hyal enzyme and generation of HA fragments (2, 43–45). Hyal type I-S and IV-S BTH were compared with rHuPH20 in our study. rHuPH20 did not stimulate reporter gene expression in TLR2- or TLR4-dependent assays, but Hyal type I-S and IV-S strongly stimulated both TLR2 and TLR4 in vitro (Fig. 4A, 4B). Consistent results were obtained with expression of MIP-2 in murine macrophages. rHuPH20 did not stimulate MIP-2 production, but Hyal type I-S and IV-S significantly stimulated MIP-2 production in Raw 264.7 cells (Fig. 4C).

We then tested the proinflammatory activity of Hyal type I-S and IV-S in vivo. All groups were treated with Hyal at an activity of 2400 U/ml. When dosed into the air pouch, both Hyal type I-S and IV-S degraded HA and caused an increase in plasma HA in a time frame similar to rHuPH20 (Fig. 5A, 5B). The size profiles of LMW HA fragments in plasma treated with Hyal type I-S and IV-S were similar to that treated with rHuPH20 (data not shown), which further confirmed the comparable in vivo Hyal activity of rHuPH20 and the Hyal type I-S and IV-S Hyals. Both Hyal type I-S and IV-S stimulated production of cytokines and chemokines in the air pouch, with profiles similar to LPS-treated samples. Hyal type I-S treatment stimulated production of MIP-1α, MIP-1β, and MIP-2 in air pouch exudates (Fig. 5C). Hyal type IV-S treatment stimulated production of IP-10, MIP-1α, MIP-1β, MIP-2, RANTES, and TNF-α in air pouch exudates.

Hyal type I-S and IV-S also induced significant neutrophil infiltration into the air pouch cavity, although to a lesser extent than LPS (Fig. 5D). Therefore, both Hyal type I-S and IV-S preparations contain proinflammatory components whereas rHuPH20 does not.

Endotoxin in bovine testis–derived type I-S and IV-S Hyals stimulates an inflammatory response in the air pouch

Endotoxin (such as LPS), a putative stimulator of acute inflammation, is a common contaminant in protein preparations (46, 47). Endotoxin levels were determined in the reagents we tested for inflammatory activity in the air pouch model, along with a summary of their activity to stimulate the innate immune response (Table I). There is a strong association between proinflammatory activity and endotoxin content. LPS at 1 μg/ml contained 3905 U/ml endotoxin; rHuPH20, Select-HA 250, and HA 200 contained very low levels of endotoxin (<1 U/ml). Hyal type IV-S contained 234 U/ml and Hyal type I-S contained 34.7 U/ml endotoxin at the concentrations used in the air pouch model. To determine whether endotoxin in BTH is a causative factor for inflammation, Hyal type I-S and IV-S were processed for endotoxin removal and subsequently tested for activity for inducing cell infiltration into the air pouch in vivo. After endotoxin removal, Hyal type I-S showed a ~50% reduction of proinflammatory activity, whereas Hyal type IV-S demonstrated a near complete reduction (Fig. 6A). Our results suggest that endotoxin is a major contributing factor in inflammation in bovine testis–derived Hyal type I-S and IV-S.

rHuPH20 is a highly purified recombinant protein preparation, whereas Hyal type I-S and IV-S are partially purified from bovine testis. Protein purity in these reagents was tested by SDS PAGE (Fig. 6B). rHuPH20 showed a single band at M_r 64,000, as previously reported by Bookbinder et al. (27). Both Hyal type I-S and IV-S showed multiple protein bands at various m.w., indicating the presence of other proteins in these reagents, which also may contribute to their proinflammatory activity. Liquid chromatography-tandem mass spectrometry analysis showed that Hyal type I-S and IV-S contain abundant contaminants in these protein preparations, including proinflammatory macrophage migration inhibitory factor, cathepsin C, and annexin A2 (Supplemental Tables I, II), whereas rHuPH20 only contains human PH20 (Supplemental Table III).

Table I. Endotoxin level in test reagents

Reagent	Endotoxin (EU/ml)	Stimulation of Innate Immune Response
PBS	<0.1	
LPS (1 μg/ml)	3905	+
rHuPH20 (20 μg/ml)	<0.1	−
Hyal type I-S (6 mg/ml)	34.7	+
Hyal type IV-S (1.6 mg/ml)	234	+
Select-HA 250 (500 μg/ml)	0.7	−
HA 200 (500 μg/ml)	<0.1	−

The endotoxin levels in the tested reagents (at the concentrations injected into the air pouch) were determined according to Endosafe-PTS Rapid Endotoxin Detection protocol (Charles River Laboratories). Elevated levels of endotoxin in LPS and Hyal type I-S and IV-S are correlated with their activity on stimulating cytokines/chemokines and neutrophil influx.

+, stimulation of innate immune response observed; −, no stimulation of innate immune response observed.

rHuPH20 inhibits inflammatory response induced by LPS, by blocking HA-CD44 interaction

HA has been reported to enhance neutrophil migration via the CD44 receptor, and reductions in polymorphonuclear leukocyte accumulation are observed in inflammatory models in CD44-deficient mice (32, 33, 48, 49). Consequently, degrading HMW HA by PH20 and other Hyals might interrupt HA–CD44 interaction and thus modulate neutrophil recruitment into inflammatory sites. To test this, we examined the effect of rHuPH20 on LPS-induced inflammation. In air pouch exudates, treatment with LPS in the presence or absence of rHuPH20 yielded similar profiles of cytokine/chemokine stimulation (Fig. 7A), indicating that rHuPH20 did not interfere with LPS-induced production of cytokines or chemokines. As expected, samples cotreated with LPS and rHuPH20 had HA degradation profiles similar to rHuPH20 alone, indicating that LPS did not interfere with the Hyal activity of rHuPH20 in vivo (Fig. 7B).

Interestingly, without impacting LPS-induced cytokine and chemokine production, rHuPH20 substantially blocked LPS-induced neutrophil infiltration into the air pouch cavity (Fig. 7C). We also examined the effect of rHuPH20 on LTA-induced (TLR2-dependent) neutrophil infiltration and found that rHuPH20 similarly inhibited neutrophil recruitment (data not shown). Our results indicate that rHuPH20 inhibits neutrophil accumulation in the air pouch, likely through disrupting HA–CD44 interaction.

Further supporting this proposed mechanism, when sections of the air pouch lining were examined microscopically, HA staining showed that HA was abundant in the air pouch lining (Fig. 8A, 8B, 8D, 8E) and was largely removed by rHuPH20 (Fig. 8C, 8F). Neutrophils in the air pouch lining were surrounded by HA (Fig. 8D, 8E, 8G, 8H) and expressed CD44 (Fig. 8G–I). Upon LPS injection, HA appeared to form a cable-like structure surrounding the neutrophils, with substantial colocalization with CD44 on the

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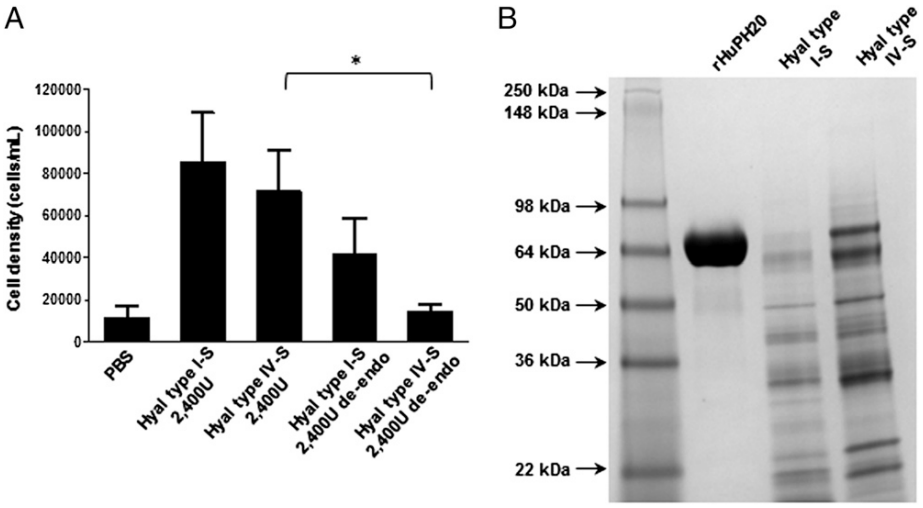

FIGURE 6. Endotoxin is a contributing factor for the proinflammatory activity of Hyal type I-S and IV-S. (**A**) Hyal type I-S and IV-S were processed for endotoxin removal using a Triton X-114 extraction, then tested in the air pouch model before and after endotoxin removal (labeled "de-endo") to determine the activity on inducing total cell influx. Results are mean readout or relative concentrations ± SD of four replicates, from one of three similar experiments. $*p < 0.05$. (**B**) rHuPH20 and Hyal type I-S or IV-S (8 μg) were separated with SDS-PAGE gel electrophoresis and stained by Coomassie blue staining. rHuPH20 had a major band at M_r 64,000, whereas both Hyal type I-S and IV-S had multiple protein bands at a wide range of m.w.

cell membrane (Fig. 8E, 8H), implying a possible role of HA–CD44 interaction in neutrophil migration. rHuPH20 removed the cable-like HA structure surrounding the neutrophils (Fig. 8F, 8I), possibly leading to inhibition of neutrophil migration into the air pouch cavity by abolishing the HA–CD44 interaction. In contrast to the air pouch cavity, LPS-induced neutrophil extravasation and subsequent accumulation in the air pouch lining was not affected by rHuPH20 (Fig. 8J–M). Therefore, rHuPH20 prevented movement of neutrophils into the cavity of the air pouch. Finally, we showed that an anti-CD44 Ab (clone IM7) injected into the air pouch concomitantly with LPS resulted in similar inhibition of neutrophil accumulation in the air pouch cavity (Fig. 8N). This supports the theory that HA–CD44 interaction is critical for neutrophil migration into the air pouch cavity.

Discussion

The cellular responses to HA of various molecular sizes and the biological impact of PH20-generated LMW HA fragments is a topic of great interest and ongoing controversy (2, 7, 14, 26). LMW HA has been reported to stimulate inflammation (24, 45, 50), but endotoxin-free HA fragments at the same molecular sizes failed to stimulate an inflammatory response in vitro (28, 29). The present study provides in vitro and in vivo evidence to demonstrate that 1) LMW HA or highly purified rHuPH20 do not stimulate inflammation and 2) in BTH preparations extracted from animal tissue, endotoxin, but not PH20, is a major contaminant that stimulates inflammation.

Although LMW HA has been reported to stimulate inflammation in a TLR2- and TLR4-dependent manner (2, 26), direct binding of

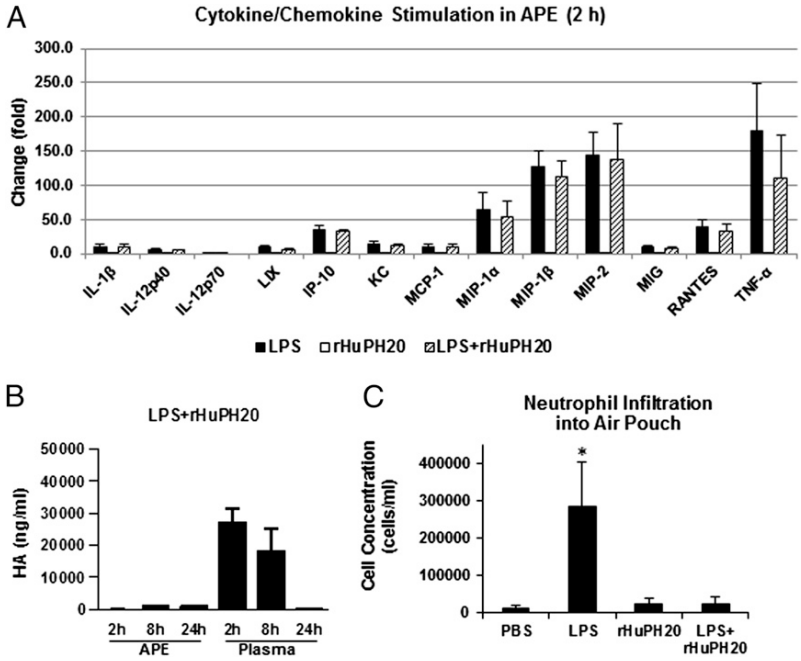

FIGURE 7. rHuPH20 inhibited LPS-induced neutrophil influx into the air pouch. (**A**) The mouse air pouch was challenged with LPS (1 μg/ml), rHuPH20 (20 μg/ml), or LPS (1 μg/ml) + rHuPH20 (20 μg/ml) for 2 and 8 h. Air pouch exudates were collected, and the protein levels of cytokines and chemokines were determined by Luminex multiplex assay ($n = 8$). (**B**) HA levels in the challenged air pouch and plasma are shown. Samples were collected as described in (A), with HA levels determined by ELISA. (**C**) Neutrophil density in the air pouch was determined by FACS analysis using anti-CD45/anti–Ly-6G in air pouch exudates collected as described in (A). Results are mean readout or relative concentrations ± SD of eight replicates from one of three similar experiments. $*p < 0.05$.

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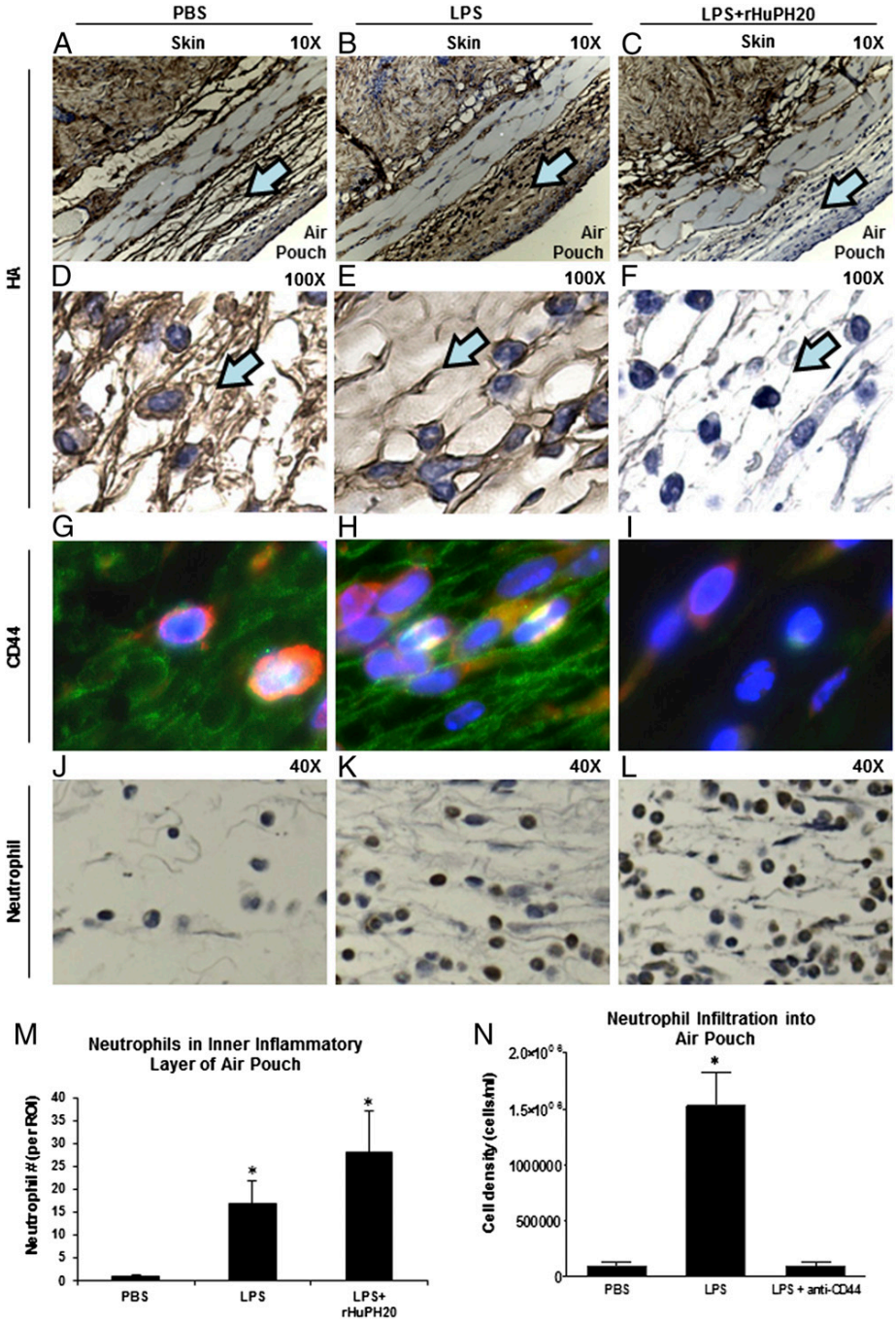

FIGURE 8. rHuPH20 inhibited LPS-induced neutrophil recruitment via blocking HA-CD44 interaction. (**A**–**F**) The air pouch skin patches challenged with PBS, LPS, or LPS + rHuPH20 were collected, stained with HA-binding protein (DAB), and micrographed. HA staining (brownish color) in air pouch linings is indicated by arrows in (A)–(C) and at high magnifications in (D)–(F). The arrows in (C) and (F) point to the location where HA was removed by rHuPH20. (**G**–**I**) Coimmunofluorescent staining of HA (green), CD44 (red), and nuclei (DAPI blue) of the treated air pouch skin patches were micrographed. (**J**–**L**) The collected air pouch skin patches were stained with anti–Ly-6G for neutrophils and micrographed. The number of neutrophils in the challenged air pouch skin patches were stained, with neutrophils in the air pouch lining counted and shown [(**M**), n = 10 fields/group]. (**N**) The mouse air pouch was challenged with PBS, LPS (1 μg/ml), and LPS + anti-CD44 Ab (500 μg/ml) for 8 h. The air pouch exudates were collected and the total influx cell density was determined by cell counting and shown (n = 4). *$p < 0.05$.

HA of any size to TLR2 or TLR4 has not yet been demonstrated. In this study, we demonstrate that LMW HA fragments do not bind or stimulate TLR2 or TLR4 on the cell surface, consistent with a recently published observation that LMW HA fragments are not TLR ligands (31). Applying LMW HA fragments to a murine macrophage cell RAW 264.7 failed to stimulate production of MIP-2, a classic proinflammatory cytokine, in agreement with previous reports that LMW HA did not stimulate production of IL-1β or NO (28, 29).

The proposed proinflammatory activity of LMW HA fragments was further tested in vivo, using a classic mouse air pouch model of inflammation (51, 52). LMW HA was reported to stimulate production of a variety of cytokines and chemokines (including MCP-1, MIP-1α, MIP-1β, RANTES, KC, MIP-2α, MIG, IP-10, IL-1β, LIX, IL-12 p40/p70, and TNF-α) and consequently induce the innate immune response (2). In the air pouch model, LMW HA (HA 200 and Select-HA 250) failed to stimulate production of any tested cytokines or chemokines. Moreover, HA 200 fragments and Select-HA 250 failed to induce neutrophil infiltration into the air pouch, suggesting that LMW HA fragments do not have proinflammatory activity in the air pouch model in vivo.

Even though Hyals have been proposed to be proinflammatory by degrading HMW HA into LMW HA fragments (2), contaminants in commonly used Hyal type I-S and IV-S make rigorous interpretation of these data problematic (31). For these reasons, we used highly purified rHuPH20 to test whether it can

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stimulate immune responses in the air pouch model. rHuPH20 administered into the air pouch did not stimulate either production of cytokines or chemokines or neutrophil infiltration into the air pouch cavity, although a significant amount of LMW HA fragments were generated in vivo. These data indicate that rHuPH20 and enzyme-generated HA fragments are not proinflammatory factors. In the air pouch model, we also tested BTH Hyal type I-S and type IV-S, the animal tissue–extracted Hyals reported to be proinflammatory (45, 50). Surprisingly, both Hyal type I-S and IV-S stimulated cytokine and chemokine production as well as neutrophil infiltration into the air pouch cavity. We hypothesized that some other components in Hyal type I-S and IV-S, but not Hyal itself, stimulated the inflammatory response in vivo.

Endotoxin, a proinflammatory factor, is a common contaminant in protein preparations and considerable caution is needed in the interpretation of mediators of inflammation, particularly when using proteins that contain the putative TLR2/TLR4-stimulating endotoxin (14, 47). It was reported that endotoxin contamination in BTH, instead of HA fragments generated by Hyal degradation, has been shown to stimulate TLRs and induce an inflammatory response (31). Indeed, we found high levels of endotoxin in Hyal type I-S and IV-S, correlating with their proinflammatory activity. No endotoxin was detected in rHuPH20 or LMW HA fragments, correlating well with their lack of inflammatory activity. Moreover, endotoxin removal from BTH (with the Hyal activity mostly retained) diminished proinflammatory activity of Hyal type IV-S and largely reduced inflammatory activity of Hyal type I-S, suggesting that endotoxin in BTH, rather than endogenous Hyal enzyme activity, stimulates the innate immune response.

Our study suggests that endotoxin removal should be a common precaution if BTH (especially Hyal type I-S and IV-S) is used in future studies to avoid artifactual stimulation of the inflammatory response. This finding has many implications for the widespread use of Hyal type I-S and IV-S BTH in tissue preparation and studies of inflammation (2, 26, 45, 50).

Neutrophil binding of HA via its surface receptor CD44 has been shown to affect multiple aspects of cellular function. HA may facilitate neutrophil accumulation at sites of inflammation at several stages. First, blockade or deficiency of CD44 results in reduced neutrophil rolling and a reduction of endothelial adhesion in vitro and in vivo (32, 33, 49). In addition, direct stimulatory effects of HA on neutrophil migration have been described in vitro (48, 53), although there are contradictory reports of HA inhibiting neutrophil migration (54, 55). These mechanisms potentially contribute to the observed inhibition of neutrophil accumulation when HA–CD44 interaction is prevented in models of inflammation such as MIP-2–induced leukocyte influx (32), renal ischemia/perfusion injury (49), and liver damage following endotoxemia (33). However, CD44 has also been implicated as a negative regulator of the inflammatory response to endotoxin in a lung model of pneumonia (56), again demonstrating the potential complexity of the role of HA in innate immunity.

We found that LPS-induced neutrophil infiltration into the air pouch cavity was dramatically inhibited by rHuPH20, which did not affect LPS-stimulated cytokine and chemokine production. This effect appeared to occur at the level of extravascular migration rather than at intravasal adhesion and endothelial transmigration, since the numbers of neutrophils in the air pouch lining were similar in animals treated with LPS alone and with LPS in combination with rHuPH20. Interestingly, in the presence of LPS the appearance of HA in the air pouch lining was altered to resemble cable-like structures. Similar findings were reported previously wherein TLR stimulation of stromal cells enhanced HA cross-linking, thereby changing its physiological structure to support leukocyte attachment (57, 58). Our studies showed that HA and CD44 were both localized on the neutrophil cell surface in the air pouch lining upon LPS treatment, suggesting a functional interaction of HA and CD44. rHuPH20 treatment removed the HA surrounding the neutrophils, interrupting the HA–CD44 interaction supposedly crucial for neutrophil migration, thus potentially preventing neutrophils in the lining membrane space from entering the air pouch cavity. Furthermore, blocking HA–CD44 interaction by anti-CD44 Ab also inhibited LPS-induced neutrophil infiltration. In both cases, the effect was strictly limited to the local extravascular space near the site of inflammation.

In summary, rHuPH20 and LMW HA fragments were studied for their potential to stimulate the innate immune response using the air pouch model of inflammation. Neither rHuPH20 nor LMW HA fragments exhibited proinflammatory activity. Bovine testis–derived Hyals (Hyal type I-S and IV-S) showed proinflammatory activity, partly because of impurities such as endotoxin. Furthermore, rHuPH20 inhibited LPS-induced neutrophil recruitment by interrupting HA–CD44 interaction, revealing a potential role of PH20 as an anti-inflammatory agent.

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Disclosures

All authors are or were employees, consultants, and/or stockholders of Halozyme Therapeutics, Inc.

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